Aliang Li*

Nominalizations and its grammaticalization in standard Thai

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Abstract: The focus of this study is on the function and grammaticalization of nominalizations in Thai and the nominalization categories and nominalizing strategies in Thai are described. The Thai language exhibits a composite of derivational and clausal nominalizations marked with three nominalizers: *kaan*^{1} derives nouns or nominalized clauses from lexical verbs and relative or complement clauses; *khwaam*^{1} derives nouns from lexical verbs and adjectives; and *kaan*^{1}*thii*^{3} is used for clauses. The current study posits diachronic developments for nominalizers and addresses related issues. It is concluded that nominalizers *kaan*^{1} and *khwaam*^{1} were originally lexical nouns meaning 'work' and 'matter'. The essential features of the use of nominalizations have remained constant, but certain developments have occurred, which include lexical nominalization to clausal nominalization and the emergence of a new nominalizer, *kaan*^{1}*thii*^{3}. It is found that nominalizers *kaan*^{1}, *khwaam*^{1} and *kaan*^{1}*thii*^{3} are basically in complementary distribution, and language internal evolution and external contact are the primary motivations for nominalization in Thai.

Keywords: Thai; derivational nominalization; clausal nominalization; nominalizer; grammaticalization

1 Introduction

Under a classic view of the architecture of the language faculty, nominalizations belong to the domain of lexical structure or morpho-syntax. Nominalization is the transformation of other word classes to nouns, and which occurs in most languages. In addition, it can also derive agentive and abstract nouns from other semantic classes of nouns. Nominalization has been a central topic of linguistic research since the in-depth analysis and discussion of the polysemous nature of nominalization in transformational grammar (Chomsky 1970; Lees 1960), and Chomsky (1970) proposed

***Corresponding author: Aliang Li**, School of Literature, Zhejiang University, Zijingang Campus, No. 866 Yuhangtang Road, Xihu District, Hangzhou, Zhejiang, China, E-mail: lialiang0908@126.com. https://orcid.org/0000-0002-8518-857X

that nominalizations are not syntactically derived but rather lexiconically derived. He divided nominalizations derived from verbal components into three types: gerundive nominals, derived nominals, and mixed forms. Since then, transformational generative grammar has taken syntactic and non-syntactic approaches to the study of nominalization (Alexiadou 2001; Borer 2003; Grimshaw 1990; Lieber 2016; Melloni 2011). In addition to transformational generative grammar, cross-linguistic or typological studies of nominalization have also received attention (Comrie 1976; Comrie and Thompson 2007 [1985]; Koptjevskaja-Tamm 1993, 2013; Gijn 2019). From a cognitive grammar perspective, nominalization involves a conceptual reification. Nominalization is pervasive while also being theoretically significant: certain of its properties are frequently taken as a means of demonstrating the autonomy of grammar and the distinction between syntax and lexicon (Langacker 1991: 22).

Nominalization construction in cross-linguistic and/or typological study is usually divided into two types: derivational/lexical nominalization and clausal nominalization (Comrie and Thompson 2007 [1985]; Yap et al. 2011). Siloni (1997) carefully defined the boundaries between the lexical nominalizing mechanism and its clausal counterpart. Derivational nominalization turns verbs and verb phrases into nouns (Comrie and Thompson 2007 [1985]; Ginnakidou and Stavrou 1999; Koptjevskaja-Tamm et al. 2015), whereas clausal nominalization nominalizes adverbial, partial complement, and relative clauses into nominalization (Coupe 2007: 11; DeLancey 2002; Genetti 2011; Noonan 1997). Clausal nominalization differs from the prototypical verbal clause and does not involve deriving nouns from verbs (nominalization is a complete clause). Verbs often retain some of their verbal properties and arguments and adjuncts have the same properties as non-nominalized clauses (Jany 2011).

Previous Thai nominalization research has focused mostly on the lexical level (Higbie and Thinsan 2002; Iwasaki and Ingkaphirom 2009; Smyth 2014), but nominalizations in Thai operate lexically and syntactically (Koptjevskaja-Tamm 2013). In this study, three common Thai nominalizers, *kaan*^{1} (Pali-Sanskrit loanword), *khwaam*^{1} (indigenous word) and *kaan*^{1}*thii*^{3} (the combination of Pali-Sanskrit and Chinese), are examined from synchronic and diachronic corpus. A synchronic corpus can comprehensively present the language facts of modern Thai, while a diachronic corpus can clearly present the function and usage of a morpheme in different historical stages. Not only can the grammaticalization process of a morpheme through diachronic data be accurately analyzed, but it can also improve credibility. The corpus used in Sections 2 and 3 is taken from *Thai National Corpus* (mainly sentences) and *Thai-English Dictionary* (mainly lexical). In Section 5 is taken from *The Princess Maha Chakri Sirindhorn Anthropology Centre Thai Inscriptions Database*, *Thai CU-concordance Database*, and *Vajirayana Digital Library*, including documents that were inscribed or written from 1283 (the date of King Ramhaeng's

stone inscription) to the present. This span of over 800 years covers three important historical periods in Thailand: (i) the Sukhothai period (AD 1238–1438) when Sukhothai was the capital for approximately 100 years; (ii) the Ayutthaya period (AD 1351–1767) when Ayutthaya was the capital for approximately 400 years; and (iii) the Bangkok period when King Rama I established Bangkok as the capital, which has lasted from AD 1782 to the present day.

The sketch outline of this study is as follows. Section 2 will first briefly review the current research status of Thai nominalization. Section 3 presents the types of nominalization in Thai from a synchronic perspective. Section 4 discusses the origin of nominalizers, while Section 5 examines the grammaticalization of nominalizers from a diachronic perspective. Finally, in Section 6, we will discuss some differences in nominalization between Thai and other Kam-Tai languages and summarize the nominalization in Thai.

2 Literature review

There are two main nominalizers in standard Thai, including morpheme *kaan*^{1} and morpheme *khwaam*^{1}. The morpheme *kaan*^{1} is a lexical noun in Thai, meaning 'work', 'affair', or 'matter', such as *kaan*^{1}*baan*^{3} (meaning 'homework'), which is a made-up word before grammaticalization. At the same time, *kaan*^{1} is also a free morpheme with word-formation ability. It can be combined with *tham*^{1} (meaning 'do') and *truuat*^{2} (meaning 'examine') to *tham*^{1}*kaan*^{1} (meaning 'do work') and *truuat*^{2}*kaan*^{1} (meaning 'examine work'). After grammaticalization, *kaan*^{1} can be used to nominalize word class (verbs), phrases, clauses, and other language constructions at different levels. Similarly, *khwaam*^{1} has developed from a lexical noun meaning 'substance' or 'sense' of a matter, fact, lawsuit, or legal case. Joseph and Janda (1988) classified this evolution from free morphemes or lexical words to affixes as the morphologization of syntax. Haas (1964: 29, 82) suggested *kaan*^{1} can be rendered in English by the suffixes *-ing*, *-tion*, *-sion*, *-al*, *-ance*, *-ence*, and *-ment*; while *khwaam*^{1} is suffixes such as *-ment*, *-th*, *-ity*, *-ness*, *- ship*, *-(t)ion*, *-ing*, *-y*, *-ery*, *-ance* (*-ence*), *-dom*, and *-ure*.

Many previous studies have researched the nominalization of Thai in recent decades. Some studies have regarded *kaan*^{1} and *khwaam*^{1} as prefixes to verbs and adjectives for converting them into abstract or verbal nouns (e.g. Campbell and Shaweevongs 1972: 201–202; Haas 1964: 29, 82; Morev 2006; Prasithrathsint 2000, 2014; Smyth 2014; Sookgasem 1996); some have suggested *kaan*^{1} and *khwaam*^{1} are nominalizers (e.g. Prasithrathsint 1996, 2005; Post 2008); and some have discussed their lexical and syntactic functions from the perspective of generative grammar (He

2011). Previous studies relating to the word class, position, and function of *kaan*[1] and *khwaam*[1] can be summarized by the fact that they can be used as nominalizers, nominalized verbs and adjectives, and referential functions.

Based on a study of the grammaticalization process of *kaan*[1] and *khwaam*[1], Prasithrathsint (1996, 2000) suggested the nominalizers *kaan*[1] and *khwaam*[1] evolved from the lexical nouns *kaan*[1] 'matter' and *khwaam*[1] 'sense' of a matter. *kaan*[1] nominalization emerged following *khwaam*[1] nominalization (see Section 5). At the same time, *kaan*[1] and *khwaam*[1] are placed before verbs and adjectives as a means of constructing morphemes. If *kaan*[1] and *khwaam*[1] are added to these words, the whole construction undergoes a syntactic transformation, the action, behavior, or nature of the word will be nominalized, and the meaning of the word may either change or remain unchanged. A nominalized construction is formed with adjective phrases and verb phrases, and the reference has the characteristics and states that are expressed by form words and verbs. The grammatical functions of *kaan*[1] and *khwaam*[1] include: (i) they have nominalized nominalizers that are integrated (nominalized) before verbs and adjectives (1); (ii) *kaan*[1]+ verb and *khwaam*[1]+verb or adjective can act as a subject (2) or object (3); (iii) nominalized relative clauses (without relativizer (4a) or with relativizer (4b)) or complement clauses (4b) in sentences.

(1) *rian*[1] 'to learn' —> *kaan*[1] (NMLZ) *rian*[1] 'learning'
təəp[2]*too*[1] 'to grow' —> *khwaam*[1] (NMLZ) *təəp*[2]*too*[1] 'growth'
dii[1] 'good' —> *khwaam*[1] (NMLZ) *dii*[1] 'goodness'

(2) a. ***kaan*[1]**-*pra*[2]*chum*[1] *tɛɛ*[2]*la*[4] *khraŋ*[4] *sin*[3]*pluaŋ*[1] *khaa*[3]*chai*[4]*caai*[2]
NMLZ-hold meeting each time cost expenses
wee[1]*laa*[1] *lɛ*[4] *plaŋ*[1]*ŋaan*[1] *pen*[1]*jaaŋ*[2]*jiŋ*[3]
Time CONJ energy in the extreme
'Each meeting costs money, time and energy in the extreme.'
(การพูดเพื่อธุรกิจ)

b. ***khwaam*[1]**-*rak*[4] *pen*[1] *ra*[4]*dap*[2] *cit*[2]*cai*[1] *suuŋ*[3] *praa*[1]*sa*[2]*caak*[2]
NMLZ-love COP level spirit REL be free
ʔa[2]*kha*[4]*ti*[2]
prejudice
'Love is a mental level that is free from prejudice.'
(*การศึกษาเชิงจิตวิทยาวินิต*)

(3) a. *khau*[5] *pen*[1]-*huaŋ*[2] ***kaan*[1]**-*khaa*[4]
3SG be-worried NMLZ-sell
'He/She is worried about business.' (*ยุทธศาสตร์เศรษฐกิจและประชาธิปไตยไทย*)

b. *khau*[5] *kam*[1]*laŋ*[1] *tham*[1] ***khwaam***[1]*-sa*[2]*aat*[2]
3SG CONT do NMLZ-clean
'He/She is doing the cleaning.' (*เจ้าไล*)

(4) a. *phom*[5] *kam*[1]*laŋ*[1] *tham*[1] *khroŋ*[1]*kaan*[1] ***kaan***[1]*-sɔɔn*[5]
1SG.MAS CONT do project NMLZ-teach
'I am working on a teaching project.' (*สุริยฉาย*)

b. *suan*[2] *kɔɔ*[1]*ra*[4]*nii*[1] ***thii***[3] *kaan*[1]*-suuu*[2]*saan*[5] *hɛɛŋ*[2]
as for case REL NMLZ=communicate LOC
pra[2]*theet*[3]*thai*[1] *mɔɔŋ*[1] ***waa***[3] ***kaan***[1]*=hai*[3] *bɔ*[2]*ri*[4]*kaan*[1]
Thailand think COMP NMLZ=give service
thoo[1]*ra*[4]*saap*[2] *khluan*[3]*thii*[3] *ra*[4]*bop*[2] *nuŋ*[2]*phan*[1]*kau*[3]*rɔɔi*[4]
telephone move system 1900
'As for the case of the Communications Authority of Thailand, it is considered that providing mobile phone service with the 1900 system.' (*นสพ.กรงเทพธรกจ 2*)

Yap et al. (2011: 3) divide nominalization construction as participant versus event, lexical versus clausal, and embedded versus non-embedded. The participant, event, and lexical nominalization belong to derivational nominalization, while embedded and non-embedded nominalization mainly belong to clausal nominalization. In addition, non-embedded (or "stand-alone") nominalizations function as complete and independent utterances, i.e. their internal structure resembles a sentence in nearly all respects. Therefore, the current study distinguishes nominalizations in Thai into three categories: derivational nominalization (including de-verbal, de-adjective, and other nominalization), clausal nominalization (embedded), and non-embedded nominalization.

3 Nominalization categories in standard Thai

Although the morphosyntactic and syntactic properties of nominalization are not as rich in Thai as in Indo-European languages or Sino-Tibetan languages, Thai nominalizations have their characteristics. The Thai language exhibits a composite of derivational and clausal nominalizations. Derivational nominalization derives nouns from lexical nouns, verbs, and adjectives. In contrast, clausal nominalization can nominalize relative clauses as the modifiers of the heads of arguments or complement clauses as the complement of a verb or adjective by adding a nominalizer at the beginning of non-embedded clauses. There are three common nominalizers in Thai, but only one is the compounding marker *kaan*[1]*thii*[3], derived from

"nominalizer + relativizer/nominalizer" by lexicalization. It should be noted that only *kaan*[1] acts as a nominalizer for derivational and clausal nominalization, which many previous studies have neglected.

kaan[1] and *khwaam*[1] are the most typical derivational nominalizers, and *khwaam*[1] is only closely combined with other verbs and adjectives without appearing between relative clauses and head nouns. *kaan*[1]*thii*[3] is a compounding nominalizer composed of the nominalizer *kaan*[1] and the relativizer/nominalizer *thii*[3]. It only appears in the syntactic construction of the clausal nominalization patterns. *thii*[3] is a typical relativizer that is mainly added between the head and modifier, such as *khruaŋ*[3]*muuu*[1] *thii*[3] *sam*[5]*khan*[1] (tool REL important). When the nominalizer *kaan*[1] and *thii*[3] are integrated, they act together as nominalizers (see Section 3.3).

3.1 Derivational nominalizations

Nominalizations are derived via nominalizer in Thai. Morphologically, we can distinguish between three types: *kaan*[1]-, *khwaam*[1]- and "other-nominalizer." Moreover, derivational nominalization is mainly divided into three types: de-verbal nominalization, de-adjectival nominalization, and other nominalization. This section starts the discussion with the first group.

3.1.1 De-verbal nominalization

De-verbal nominalization refers to the nominalization of abstract actions. Due to different metaphorical and metonymic processes, it can refer to objects, emotions, people, action times, and places (Aidsa 2006). The nominalizer *kaan*[1] is added before verbs, particularly action verbs, and nouns, as a means of making them abstract, which indicates the corresponding abstract concept of the action or thing.

kaan[1]	+	*nam*[1] 'to lead'	—>	*kaan*[1]*nam*[1] 'leading'
kaan[1]	+	*pra*[2]*moŋ*[1] 'to fish'	—>	*kaan*[1]*pra*[2]*moŋ*[1] 'fishing'
kaan[1]	+	*ban*[3] 'house'	—>	*kaan*[1]*ban*[3] 'housework'
kaan[1]	+	*ŋan*[1] 'work'	—>	*kaan*[1]*ŋan*[2] 'affair'

The nominalizer *khwaam*[1] is used for non-action verbs and verbs that express states, feelings, natures, and attributes are transformed into abstract nouns.

khwaam[1]	+	*ruu*[4] 'to know'	—>	*khwaam*[1]*ruu*[4] 'knowledge'
khwaam[1]	+	*waŋ*[5] 'to hope'	—>	*khwaam*[1]*waŋ*[5] 'hope'
khwaam[1]	+	*khit*[4] 'to think'	—>	*khwaam*[1]*khit*[4] 'thinking'

There is often a blurring of the boundaries between the use of the Thai nominalizers *kaan*[1] and *khwaam*[1] in the de-verbal nominalization. Thus, the distinction between the two has been discussed (e.g. Prasithrathsint 1996, 2005; Morev 2006). Prasithrathsint (2005) suggested that some non-action verbs should be nominalized with *khwaam*[1], and it was found that they are to be nominalized with *kaan*[1]:

rap[4]*ruu*[4]	'to perceive'	*kaan*[1]*rap*[4]*ruu*[4]	* *khwaam*[1]*rap*[4]*ruu*[4]	'perception'
dai[3]*jin*[1]	'to hear'	*kaan*[1]*dai*[3]*jin*[1]	* *khwaam*[1]*dai*[3]*jin*[1]	'hearing'

Also, some verbs are found to occur with both *kaan*[1] and *khwaam*[1]:

klom[1]*klɯɯn*[1]	'to assimilate'	*kaan*[1]*klom*[1]*klɯɯn*[1]	'assimilation'
		khwaam[1]*klom*[1]*klɯɯn*[1]	'harmony'
ruam[3]*mɯɯ*[1]	'to cooperate'	*kaan*[1]*ruam*[3]*mɯɯ*[1]	'cooperation'
		khwaam[1]*ruam*[3]*mɯɯ*[1]	

Prasithrathsint (2005) proposed three categories of Thai verbs for distinguishing *kaan*[1] and *khwaam*[1] as nominalizers: perceptible, imperceptible, and balanced verbs. The study found that perceptible verbs occur with *kaan*[1] and non-perception verbs occur with *khwaam*[1]. Verbs in the third category were found to be the most problematic as they are ambivalent regarding their ability to occur with either *kaan*[1] or *khwaam*[1]. Generally speaking, the choice of nominalizer is based on semantic considerations in all cases.

3.1.2 De-adjectival nominalization

This type of nominalization is mainly dependent on the addition of *khwaam*[1] before an adjective as a means of changing the adjective into a noun with the nature of an adjective. The nominalized adjective freely acts as the topic of the copula clause (5), subject (6), and object (7) in a sentence.

(5) *sam*[5]*lap*[2] ***khwaam***[1]*-jaau*[1] *ʔɯɯn*[2]*-ʔɯɯn*[2] ***nan***[4] *kɔ*[3] *jaŋ*[1]
for NMLZ-long other-REDUP TOP CONN still
pen[1] *raai*[1]*kaan*[1] *thɛɛm*[5] *fe*[1]*rii*[1] *ʔiik*[2]
COP item add free another
'For other lengths, they are also free items.' (*ทฤษฎีให้บริการปรึกษา*)

(6) ***khwaam***[1]*-cha*[2]*laat*[2] *pen*[1] *sa*[2]*nee*[2] *jaaŋ*[2]*nɯŋ*[2] *khɔɔŋ*[5] *ma*[4]*nut*[4]
NMLZ-intelligent COP charm for one thing GEN human
'The intelligence is one of the charms of humans.' (*กบเหลาดินสอ*)

(7) *mai³pen¹rai¹ rɔɔk² phom⁵ mii¹* ***khwaam¹-suk²*** *dii¹*
that is all right NEG.PRT 1SG.MAS have NMLZ-happy good
'It is ok. I am well-being.' (*สวัสดีข้างถนน*)

3.1.3 "Other nominalization"

Under this label, we include deverbal agent nominalizations and agent nominalizations of nouns derived by various nominalizers (*nak⁴*-, *chaaŋ³*-, and *thii³* (regarding *thii³*, see Section 4.2)). The first is deverbal agent nominalizations. The nominalizers *nak⁴*- and *chaaŋ³*- added before the verb transform the action verb into the agent of the action, denoting the person who engages in the behavioral counterpart of that action. This process nominalizes the verb into the nominal argument of the verb, i.e. the agent, and is a productive mechanism. The referents after nominalization are the emitter of the action, a typical agent nominalization (Noonan 1997: 375). The agentive nominalizer creates agentive nominals where the person involved normally does the action as a job or regular activity (LaPolla 2008: 50).

nak⁴	+	*khian⁵*	'to write'	—>	*nak⁴khian⁵*	'writer'
nak⁴	+	*te²*	'to kick'	—>	*nak⁴te²*	'footballer'
nak⁴	+	*bin¹*	'to fly'	—>	*nak⁴bin¹*	'aviator'
chaaŋ³	+	*kɔɔ²saaŋ³*	'to build'	—>	*chaaŋ³kɔɔ²saaŋ³*	'builder'
chaaŋ³	+	*chɯam³*	'to weld'	—>	*chaaŋ³chɯam³*	'welder'
chaaŋ³	+	*phim¹*	'to print'	—>	*chaaŋ³phim¹*	'printer'

Langacker (1987: 445–446) chose the [V-er] nominalization pattern for illustrating the alternate profiling nominalization. The stem and nominalizer are component constructions that are integrated to form the composite construction (N) (Langacker 1991: 23). This semantic schema of such nominalization in Thai is shown in Figure 1.

The agent nominalizations of nouns refer to the nominalization of event or thing nouns into agent or experiencer with their property and characteristics. *nak⁴*- or *chaaŋ³*- can be used for such nominalizations.

nak⁴	+	*kii¹laa¹*	'sport'	—>	*nak⁴kii¹laa¹*	'athlete'
nak⁴	+	*khaau²*	'news'	—>	*nak⁴khaau²*	'reporter'
nak⁴	+	*khee¹mii¹*	'chemistry'	—>	*nak⁴khee¹mii¹*	'chemist'
chaaŋ³	+	*ŋəən¹*	'silver'	—>	*chaaŋ³ŋəən¹*	'silversmith'
chaaŋ³	+	*mai⁴*	'wood'	—>	*chaaŋ³mai⁴*	'carpenter'
chaaŋ³	+	*ʔaa¹kaat²*	'air'	—>	*chaaŋ³ʔaa¹kaat²*	'aircraftman'

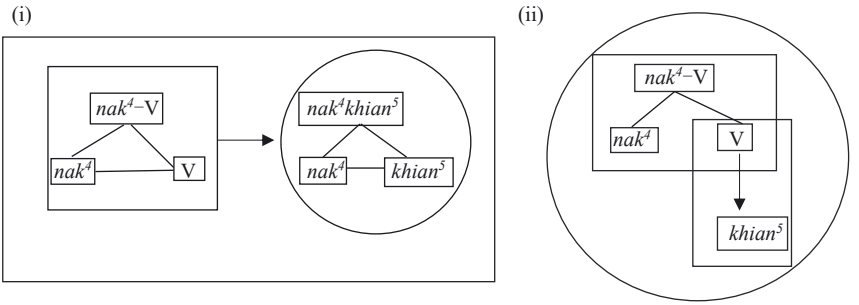


Figure 1: The two different representations of the [*nak*^{4}-*chaaŋ*^{3}-V] pattern of agentive nominalization in Thai. In diagram (i), the schematic unit is shown on the left. The component structures [*nak*^{4}-] and [*khian*^{5}] integrate to form the composite structure [*nak*^{4}-*khian*^{5}]. Diagram (ii) depicts the transition chain underlies the actualization of [*nak*^{4}-*khian*^{5}].

3.2 Clausal nominalizations

"When suffixed to a clause, the nominalizer forms a relative or complement clause that functions as an adjunct, for instance to a noun, or an argument of a verb" (Doornenbal 2009: 87). The nominalizer *kaan*^{1} is used in two ways in combining clauses, while *kaan*^{1}*thii*^{3} is used in one way. The general nominalizer as a subordinator:

i. forms an adnominal or nominal phrase, i.e., a relative clause by relativization or preposition;
ii. forms a sentential complement.

3.2.1 Relative clauses

Yap et al. (2011: 3) studied nominalizers that may go on to derive adjectives and relative clauses. In this section, we will follow Andrews' (2007: 206) definition of relative clauses from a typology perspective. "A relative clause (RC) is a subordinate clause which delimits the reference of an np by specifying the role of the referent of that np in the situation described by the RC." Furthermore, Andrews (2007: 208) divides relative clauses into two main types based on Hale (1976) and Andrews (1985: 11): embedded relative clauses and adjoined relative clauses. The former can also be divided into three subcategories based on the position of domain nominal (or head noun), i.e. external, internal, and free (embedded) relative clauses.

Thai permits embedded relative clauses to be marked, and the semantic relationship between the domain nominal and its relative clause is lexically specified. Most nominalized clauses can function as NPs of prepositional phrases in Thai, which

are limited in form and marked by prepositions *nai*[1] and *sam*[5]*rap*[2]. These prepositional phrases act as post attributes to modifier domain nominal (*pra*[2]*sop*[2]*kaan*[1] 'experience' and *wi*[4]*thii*[1] 'way') and function as the equivalent of relative clauses (8a) and (9). Nominal attributes modify their noun phrase heads, i.e. nominalized attributes that function as relative clauses (Coupe 2007: 208). We argue that in such sentences, the prepositions serve to connect the head noun to the nominalized clause, which is equivalent to a relativizer-like (*nai*[1] in (8a) and *sam*[5]*rap*[2] in (9)). The nominalizers are limited in *kaan*[1] in such relative clauses. A nominal attribute derived by the preposition *nai*[1] or *sam*[5]*rap*[2] + nominalizer *kaan*[1] functions as a relative clause. In addition, the nominalized clause can directly modify relative NP without needing a relativizer (8b). Relative clauses that precede the domain nominal are more likely to be nominalized than those that follow but do not appear to use special "relative pronouns" to express NPrel function, tending rather to lack overt NPrel (Andrews 2007: 208–209). Although (8b) is a relative clause that follows the domain nominal, it does not appear to use special 'relative pronouns' as well.

(8) a. [*pra*[2]*sop*[2]*kaan*[1] **[*nai*[1] *kaan*[1]**=*rian*[1] *phaa*[1]*saa*[5] *thai*[1]]] *mii*[1]
experience PREP NMLZ=learn language Thai have
jə[4] *maak*[3]
much very
'There is much experience in learning Thai.'
(*ภาษาในสังคมไทย*)

b. [*pra*[2]*sop*[2]*kaan*[1] [***kaan*[1]**=*rian*[1] *phaa*[1]*saa*[5] *thai*[1]]] *mii*[1]
experience NMLZ=learn language Thai have
jə[4] *maak*[3]
much very
'There is much experience in learning Thai.'

(9) [*wi*[4]*thii*[1] **[*sam*[5]*rap*[2]** ***kaan*[1]**-*dəən*[1]*thaaŋ*[1]]] *saa*[5]-*maat*[3] *chuai*[3] *nak*[4]
way PREP NMLZ-tour able-to help expert
thɔɔŋ[3]*thiau*[3] *dai*[3]
tour can
'Travel guides can help tourists.'
(*การจัดการการเกษตรทรัพยากรธรรมชาติและการท่องเที่ยว*)

Firstly, it should be noted that there are embedded relative clauses in Thai, which is in line with the fact that the relative clause is behind the domain nominal and plays a modifying and limiting role. These relative clauses are external because external RCs follow the domain nominal. Andrews (2007: 233) claimed that nominalization appears to be restricted to embedded RCs. In fact, (8a) and (9) are not proper relative clauses but preposition + nominalization constructures. A

preposition + nominalization structure appears behind the domain nominal to form "NP+PP" (head noun + modifier), resulting in a relative clause and a relativizer-like. Let us look at (8b), where (8b) is a clausal nominalization directly modifying the domain nominal, forming "NP+NP" (head noun + modifier). The nominalizer *kaan*[1] in (8b) is located between the domain nominal and the nominalized constituent. Thus, the nominalizer serves as a regular noun relativizer, embedding a clause in a noun phrase as a modifier to the noun. The structures of the relative construction are as follows:

(10) a. [...N $_{\text{domain nominal}}$...]-PREP+NMLZ $_{\text{relative clause}}$
b. [...N $_{\text{domain nominal}}$...]-NMLZ $_{\text{relative clause}}$

In addition, the subject is modified and restricted by prepositional phrases acting as relative clauses (11), which can be topicalized. In contrast, prepositional phrases can act as attributes to modify and restrict the object (12). According to Watters (2008), nominalizations such as (12) can be called non-subject nominalizations or non-subject relative clauses.

(11) [*khwaam*[1]*-chuai*[3]*lɯɯa*[5] **[*nai*[1] *kaan*[1]**=*cak*[2] *khlooŋ*[1]*kaan*[1]]] ***nan***[4]
NMLZ-help PREP NMLZ=split project TOP
dai[3] *kha*[2]*jaai*[5] *khɔɔp*[2]*kheet*[2]
COMP expand scope
'Help in recognizing that the project has expanded its scope.'
(*รวมคอลัมน์ไม่มีผู้แต่งนิตยสารดิฉัน*)

(12) *mai*[3] *mii*[1] *sa*[2]*thaa*[5]*nii*[1] *dai*[1] *pra*[2]*sop*[2] [*khwaam*[1]*-sam*[5]*ret*[2]
NEG have station INDEF achieve NMLZ-successful
[***nai***[1] ***kaan***[1]=*soŋ*[2] *phaan*[2] *freem*[1] *khɔɔ*[3]*muun*[1]]]
PREP NMLZ=send through frame data
'No station was able to transmit data frames successfully.'
(*การสื่อสารข้อมูล*)

3.2.2 Complement clause

Complementation is the complement of a verb or adjective and often has a connection to the subject by a linking verb or other word class. This linking verb is known as a complementizer, and it can replace the narrative role of the verb and make up the sentence to obtain a whole meaning. The Thai verb *waa*[3], i.e. speech act verb *waa*[3], functions as a complementizer *waa*[3]; it can be seen that *waa*[3] no longer plays the role of the main sentence verb and can be combined with verbs, conjunctions, and copulas with the semantics of "perception" (13a), "cognition" (13b), and

"expression" (13c). The structure of X+(±modifier) +*waa*^{3}+clause, where the clause can be a complement clause. A complementizer can introduce and mark a complement clause. A complement clause "involves objectifying or reifying a complete sentential proposition such that it may occupy the functional slot of an NP within a larger sentence" (Herring 1991: 63). In Thai language, it appears that complement clauses use the same nominalizer found in relative clauses and another one. Therefore, when a complement clause is a verb complement clause, it should be nominalized by *kaan*^{1} (13) or *kaan*^{1}*thii*^{3} (14).

(13) a. *khau*^{5} *klua*^{1} ***waa*^{3}** ***kaan*^{1}**=*chai*^{4} *chii*^{1}*wit*^{4} *jaaŋ*^{1}
3SG afraid COMP NMLZ=spend life ADV
lam^{1}*baak*^{2}
painfully
'He/She was afraid (that) life would be difficult.' (*ลำเนาลม*)

b. *rat*^{4}*tha*^{2}*baan*^{1} *ra*^{4}*bu*^{2} ***waa*^{3}** ***kaan*^{1}**=*pɔɔt*^{2}*kwuaŋ*^{3} *suu*^{2}
government mention COMP NMLZ=open towards
phaai^{1}*nɔɔk*^{3}
outside
'The government says it will open to the outside world.'
(*รวมคอลัมน์เขียนทองนิตยสารพลอยแกมเพชร*)

c. *phom*^{5} *jaak*^{2} *ruu*^{4} ***waa*^{3}** ***kaan*^{1}**=*mii*^{1} *raaŋ*^{3}*kaai*^{1}
1SG.MAS want know COMP NMLZ=have body
kam^{1}*not*^{2} *cit*^{2}*cai*^{1} *phom*^{5} *ruuu*^{5} *cit*^{2}*cai*^{1} *phom*^{5}
limit mind 1SG.MAS or mind 1SG.MAS
kam^{1}*not*^{2} *raaŋ*^{3}*kaai*^{1}
limit body
'I want to know if having a body determines my mind or my mind determines my body.' (*ตามหาข้าพเจ้า*)

(14) *phom*^{5} *kɔ*^{3} *hen*^{5} *waa*^{3} ***kaan*^{2}-*thii*^{3}**=*mii*^{1} *nuui*^{2}*ŋaan*^{1}
1SG.MAS also think COMP NMLZ-REL=have agency
ʔuun^{2} *ʔɔɔk*^{2}*maa*^{1} *khluan*^{3}*wai*^{5} *thuuu*^{5} *pen*^{1} *ruuŋ*^{3} *dii*^{1}
other come out move consider COP thing good
'I also thought that having other agencies come out and take action is a good thing.' (*นสพ.กรงเทพธรกจ 2*)

(13) and (14) belong to the nominalization of the object complement, and the subject complement can likewise be nominalized. As was previously mentioned, the complement occupies an NP position and should be nominalized if the complement is a verb complement clause in (15a) with *kaan*^{1} and (15b) with *kaan*^{1}*thii*^{3}. In addition, the nominalized clauses in (15) function as the complement of the copula *khuuu*^{1}, and

nominalizers *kaan*[1] and *kaan*[1]*thii*[3] convey the sense of certainty about the proposition expressed by the nominalized clause. Nominalized clauses that are embedded into an equative are emphatic constructions that are "used to assert the speaker's strong belief in the truth value of a proposition" (Genetti 1994: 165).

(15) a. *khrooŋ*[1]*ruaŋ*[3] *lak*[2] ***khuu***[1] ***kaan***[1]=*tham*[1] *khloon*[1] *ma*[4]*nut*[4]
plot main COP NMLZ=do clone human
nai[1] *ʔa*[2]*naa*[1]*khot*[4]
LOC future
'The main plot is to clone humans in the future.'

b. *pra*[2]*theet*[3] *nii*[4] *mii*[1] *chuu*[3] *ʔiik*[2] *ruaŋ*[3] *nuŋ*[2] *kɔ*[3]
country DEM have name another thing one CONN
khuu[1] ***kaan***[2]***-thii***[3]=*mii*[1] *tha*[4]*naa*[1]*khaan*[1] *thii*[3] *khrai*[1] *mii*[1]
COP NMLZ-REL=have bank REL REFL have
ŋəən[1] *kɔ*[3] *saa*[5]*maat*[3] *pai*[1] *pəət*[2] *ban*[1]*chii*[1] *lap*[4]
money CONN able to go open account secret
dooi[1] *mai*[3] *hai*[3] *khrai*[1] *ruu*[4] *dai*[3]
INSTR NEG CAUS REFL know can
'Another thing this country is famous for is that it has banks where anyone with money can go and open a secret account without anyone knowing.' (*นสพ.กรุงเทพธรกจ 2*)

3.3 Non-embedded nominalization

Nominalization constructions can be either lexical or clausal nominalizations. Clausal nominalization can be subdivided into embedded and non-embedded nominalization. Non-embedded clauses can be nominalized as a means of expressing the position of the speaker (Bickel 1999; DeLancey 2002; Watters 2008; Yap and Matthews 2008). Non-embedded nominalizations have complete and independent functions (Yap et al. 2011: 7). Many previous studies suggested that a major function of stand-alone nominalizations is for setting and marking background events (Rutgers 1998: 235, 247; Opgenort 2004: 325). In other words, it can act as "backgrounding", something that is "not part of the progressing story" (Doornenbal 2009: 204).

Non-embedded is nominalized by *kaan*[1]*thii*[3] in Thai. *kaan*[1]*thii*[3] consists of nominalizer *kaan*[1] and relativizer *thii*[3]. Some of the functions of *kaan*[1]*thii*[3] are embodied in its nominalization construction. *kaan*[1]*thii*[3] occurs at the beginning and is an emphatic marker described by the sentence. Through sentence-initial *kaan*[1]*-thii*[3], the speaker commits to the fact of what is said.

(16) ***kaan²-thii³***=*mii¹* *nak⁴rian¹* *pai¹* *rian¹* *don¹trii¹* *nai¹*
NMLZ-REL=have student go study music LOC
taaŋ²pra²theet³ *phrau⁴khau⁵* *tɔɔŋ³kaan¹* *pai¹* *haa⁵* *khruu¹* *thii³*
foreign country because need go search teacher REL
dii¹
good
'Having students study music abroad because they need to find a good teacher.' (*นสพ.กรงเทพธรกจ 2*)

kaan¹thii³ can also co-occur with a topic marker *nii⁴* or *nan⁴* as a means of forming a frame construction. This construction emphasizes mood and the resultant nominalized clause can be a topic, as is demonstrated in (17). This means that the presence of the topic marker *nan⁴* after *kaan¹thii³* strongly suggests that *kaan¹thii³* has the function of a sentential nominalizer. The compounded *kaan¹thii³* and *nan⁴* can be the so-called "repackaging of information" and this results in structural reanalysis and structural change, as was discussed extensively by Roberts and Roussou (2003). The speaker in (17a) wishes to emphasize the importance of expatriate ambassadors and focuses on the activity of expatriate ambassadors.

(17) a. ***kaan¹-thii³***=*soŋ²* *ʔeek²ʔak²khra⁴raat³cha⁴thuut³* *pai¹*
NMLZ-REL=send ambassador go
pra²cam¹jaŋ¹ *taaŋ²pra²theet³* ***nan⁴*** *tham¹hai³* *taaŋ¹pra²theet³*
consul foreign-country TOP CAUS foreign-country
dai³ *ruu⁴cak²* *pra²theet³thai¹* *jiŋ³* *khun³*
can know Thailand more DIR
'Sending ambassadors to foreign countries will make foreign countries know Thailand more and more.' (*นสพ.กรงเทพธรกจ*)

b. ***kaan¹***=*soŋ²* *ʔeek²ʔak²khra⁴raat³cha⁴thuut³* *pai¹* *pra²cam¹jaŋ¹*
NMLZ=send ambassador go consul
taaŋ²-pra²theet³ ***nan⁴*** *tham¹hai³* *taaŋ¹pra²theet³* *dai³*
foreign-country TOP CAUS foreign-country can
ruu⁴cak² *pra²theet³thai¹* *jiŋ³* *khun³*
know Thailand more DIR
'Sending ambassadors to foreign countries will make foreign countries know Thailand more and more.'

kaan¹thii³ is used in (17a). The sentence is focused on the expatriate ambassador due to the fact that it predicts the outcome of being an ambassador abroad. In this sentence, *kaan¹thii³* is used solely for focusing on the state that is forecast by the sentence, i.e., emphasizing the state. At the same time, the nominalizer *kaan¹* is used in (17b). The sentence describes the completion of the action *soŋ²* 'send' and there is

particular emphasis on the consequent state of the action, which is the fact that sending ambassadors to foreign countries will help them better understand Thailand. It is believed that *kaan*^{1} can replace *kaan*^{1}*thii*^{3} in certain cases and also mark a sentence into a nominalized clause. The clausal nominalization that is guided by *kaan*^{1} should omit *thii*^{3} based on *kaan*^{1}*thii*^{3}, thereby aligning with the economic principle of language.

4 The origin and development of nominalizers

The previous discussion found there to be three nominalizers in Thai: *kaan*^{1}, *khwaam*^{1}, and *kaan*^{1}*thii*^{3}. *kaan*^{1} and *khwaam*^{1} can be derivational nominalizers, while *kaan*^{1}*thii*^{3} cannot. However, *kaan*^{1} and *kaan*^{1}*thii*^{3} can also be used for connecting embedded and non-embedded clauses as nominalizers of typical clause patterns. It should be noted that in all the types of nominalization that are mentioned above, *kaan*^{1}*thii*^{3} appears in the syncretic form and it is treated as the attachment construction of [*kaan*^{1}+*thii*^{3}] in the grammatical annotation. It is common for many Kam-Tai languages to use a nominalizer as a relativizer, but why does the Thai language use a grammatical marker *kaan*^{1}*thii*^{3}, which is a combination of nominalizer *kaan*^{1} and relativizer *thii*^{3}, to perform nominalization? Some nominalized constructions can be combined with *kaan*^{2}, but why are they combined with *thii*^{3}? From the above, it can be seen that *kaan*^{1} is a nominalizer. This study investigates the origin and related development of *kaan*^{1}, *khwaam*^{1}, and *kaan*^{1}*thii*^{3} in an attempt to solve these problems.

4.1 Pali-Sanskrit loanwords

From an etymological perspective, *kaan*^{1} in Thai is a Pali-Sanskrit loanword that can derive new words after borrowing from Thai. According to Thai scholar Thongprasoet (2003: 294), *kaan*^{1} is derived from the Pali-Sanskrit **กรฺณ** with the stem **กรฺ** [*ka*^{2}*ra*^{2}]+the suffix **ณ**. As the short vowel /a/ in **ก** is made longer by suffix **ณ**, it becomes a long vowel /aa/, so **กรฺ** [*ka*^{1}*ra*^{1}] becomes **การ** [*kaan*^{2}]. The stem **กรฺ** [*ka*^{1}*ra*^{1}] is a verbal stem, which is equivalent to 'to do'.

According to *The Royal Institute Dictionary* (Royal Institute of Thailand 2011) compiled by the Royal Society of Thailand, the definition of *kaan*^{1} is as follows: it can be used as the suffix of compound words in Pali and Sanskrit (18), and it serves as the suffix of word-formation in Pali and Sanskrit (19).

(18) *nak*^{4} + *kaan*^{1} *nak*^{4}*kaan*^{1} postman
phuu^{3}*-baŋ*^{1}*khap*^{4} + *kaan*^{1} *phuu*^{3}*-baŋ*^{1}*khap*^{4}*kaan*^{2} commander

(19) *raat*^{1}*cha*^{4} + *kaan*^{1} *raat*^{1}*cha*^{4}*kaan*^{1} public sevice
phaa^{1}*nit*^{3}*cha*^{3}*ja*^{3} + *kaan*^{1} *phaa*^{1}*nit*^{3}*cha*^{3}*ja*^{3}*kaan*^{1} government service

kaan^{1} is generally considered to be a Pali-Sanskrit loanword, whereas *khwaam*^{1} is indigenous Thai word not a loanword from another language. Morev (2006) suggested that *khwaam*^{1} to be an indigenous word with a vague meaning of 'matter', 'case', or 'statement'. It can be argued that *khwaam*^{1} is an indigenous Thai word through an examination of the language of its Thai relatives. There is a Tai Lue equivalent of *khwaam*^{1} in Chinese, *xwaam*^{2}, and Thai *khwaam*^{1} and Tai Lue (Xishuangbanna Dai language) *xwaam*^{2} are Thai language cognates. When Tai Lue *xwaam*^{2} is used as a noun, its meaning is the same as Thai *khwaam*^{1} (20), which means 'meaning', 'matter', or 'substance'. When Tai Lue *xwaam*^{2} is used as a nominalizer, it serves the same function as Thai *khwaam*^{1} (21).

(20) noun
Thai: *khwaam*^{1} 'thought', 'substance'
Tai Lue: *xwaam*^{2} 'thought', 'substance'
(Luo 1988)

(21) nominalizer
Thai: *khwaam*^{1}*+fan*^{5} 'NMLZ-to dream' —> 'dream'
Tai Lue: *xwaam*^{2}*+fan*^{1} 'NMLZ-to dream' —> 'dream'
(Luo 1988)

4.2 The correlation between nominalization and relative clause

Thai nominalizer *kaan*^{1} and relativizer *thii*^{3} are integrated into an inseparably compounding nominalizer *kaan*^{1}*thii*^{3}. Comrie and Thompson (2007) suggested that one of the functions of nominalization is modifying the head noun as a relative clause. In some languages, distinguishing between relativizer and nominalizer can be difficult, and Asian languages generally use nominalizer and relativizer in the same form, such as *de* in Chinese. *thii*^{3} in Thai generally acts as a relativizer in sentences and an abundance of research has been conducted on this (Huang and Jenks 2013; Jenks 2011; Kuno and Wongkhomthong 1981). In addition, previous studies showed that the Sinitic etymology of *thii*^{3} is thought to be borrowed from Middle Chinese 地 *dijH*, meaning 'place' (Bisang 1996; Pittayaporn 2014; Prapin 1975). Kuno and Wongkhomthong (1981: 195) also suggested that the "original meaning is 'place' and which is also as a locative preposition meaning 'at'."

Thai *thii*^{3} is a complementizer and a relative operator (22). In addition to appearing before the relative clause (22a), it appears before the noun complement clause (22b) (Jenks 2011: 138).

(22) a. *nak*^{4}*riian*^{1} ***thii***^{3} *khruu*^{1} *khuan*^{1} *tii*^{1} *son*^{2} *maak*^{3}
student REL teacher MOD hit naughty very
'The student that the teacher should hit is very naughty.'

b. *chan*^{5} *mai*^{3} *chɔɔp*^{3} *khwaam*^{1}*-khit*^{4} ***thii***^{3} ***waa***^{3}
1SG.NEUT NEG like NMLZ-think REL.COMP COMP
khruu^{1} *khuan*^{1} *tii*^{1} *nak*^{4}*rian*^{1}
teacher MOD hit student
'I do not like the idea that teachers should hit students.'
(Jenks 2011:139)

Thai *thii*^{3} appear in an appropriate syntactic environment, first in the relative clause (22a) and then the noun-complement clause (22b). In both environments, *thii*^{3} is required. If it is supposed that *thii*^{3} is a complementizer (22b), then the simple statement in (22b) that *thii*^{3} is a complementizer is sufficient. However, in a noun-complement clause, *thii*^{3} is optionally followed by another complementizer, *waa*^{3}. The extra word COMP is omitted as it also introduces the complement of the verb, from which *thii*^{3} is absent in (13a)–(13c).

thii^{3} can be used as a nominalizer in front of some verbs (23) or clauses (24) in Thai. The former is derivational nominalization, and nominalizer *thii*^{3} acts as the locative (23a), instrument (23b), and agentive nominalizer (23c). Bisang (1996) suggested that *thii*^{3} in (23) is a class noun. The nominalizer *de* in Chinese is representative in this respect. Zhu (1982: 77) noted that in modern Chinese, the primary means of transforming the predicate component into the substantive component is adding an auxiliary word to the predicate part of the word as a means of creating word structure. For example, *youyong* 'swim' originally refers to an action, and with the addition of *de*, *youyong-de* 'swimmer' refers to the person who acts. The function of nominalization of Thai *thii*^{3} is similar to that of Chinese *de*. However, it has a limited scope, so it is not as universal and analogical as other Thai nominalizers, *kaan*^{1} and *khwaam*^{1}.

(23) a. *thii*^{3} + *juu*^{2} 'to live' —> *thii*^{3}*juu*^{2} 'address'
b. *thii*^{3} + *naŋ*^{3} 'to sit' —> *thii*^{3}*naŋ*^{3} 'seat'
c. *thii*^{3} + *rak*^{8} 'to love' —> *thii*^{3}*rak*^{8} 'darling'

(24) ***thii***^{3}*=khun*^{1} *phuut*^{3} *maa*^{1} ***nan***^{4} *phom*^{5} *hen*^{5}*duuai*^{3}
NMLZ=2SG speak PFV TOP 1SG.MAS agree
'I agree with what you said.'

In summary, many studies have shown that Thai *thii*^{3} has undergone a grammaticalization process [*thii*^{3} N 'place'>CLASS>COM/REL>NOM/PREP>CLF] (Kullavanijaya 2008: 463; Loss 2017: 133). Regarding historical evolution, nominalization is an essential part of the continued development of relativization and this is reflected in several languages where relativizers are closely related to nominalizers (Jany 2011; LaPolla 2008). The compounding nominalizer *kaan*^{1}*thii*^{3} in Thai is derived from the functional analogy between *kaan*^{1} and *thii*^{3} acting as nominalizers. *kaan*^{1} and *thii*^{3} can also be used together as they can nominalize clauses, which is consistent with the function of the compounding nominalizer *kaan*^{1}*thii*^{3} for nominalizing (non) embedded clauses. "Compounding is a reanalysis that involves the weakening and often loss of the boundary between words or morphemes and sometimes, the result is a derivational morpheme" (Hopper and Traugott 2003: 58).

5 The grammaticalization process of nominalizers

Himmelmann (2004: 25) noted that "grammaticization or grammaticalization may distinguish two basic usage types: a very broad and unspecific one in which the term means 'be/become part of grammar', and a narrower and more specific one where it refers to processes leading towards greater grammaticality of linguistic items. The broad usage does not necessarily refer to historical processes, nor does it presuppose that one is dealing with some kind of a directional development." However, the narrow usage does precisely that, explicitly referring to directional change processes over time. In this paper, the object is limited to narrow grammaticalization. Bybee (2010: 105) suggested that "synchrony and diachrony have to be viewed as an integrated whole; change is both a window into cognitive representations and a creator of linguistic patterns." From their perspective, there must be some correspondence between synchronic variation and diachronic evolution. The study of grammaticalization and diachronic development of Mainland Southeast Asia as a whole has received much attention (Bisang 1996 and other). Siebenhütter (2020: 78) suggested that "contact-induced linguistic transfer trigger borrowings and replications of meaning and linguistic structures." She further explains some of the grammaticalization in Thai in terms of historical factors and language contact, with Thai borrowing a large number of words from Sanskrit-Pali, which is the Buddhism languages. Bisang (2021: 773) points out "grammaticalization is a widespread phenomenon in Mainland Southeast Asian languages, and reveals two corollaries of pragmatic inference: (i) The retention of earlier meanings blocks layering and supports the accumulation of meaning on individual linguistic signs. (ii) The emergence of specific syntactic patterns (attractor positions) potentially operates against the gradualness of grammaticalization."

The Thai nominalizers *kaan*[1], *khwaam*[1], and *kaan*[1]*thii*[3] have different origins and emergence periods. The period of appearance of *kaan*[1] and *khwaam*[1] is relatively straightforward. Prasithrathsint (1996, 2005) noted that three incidences of *khwaam*-nominalization are evident from the Sukhothai period (the 13th century). The *kaan*-nominalization, which is the most general pattern that exists today, emerged during the late Ayutthaya period (17th century), but became productive in the 20th century. However, when *kaan*[1]*thii*[3] came into being must be clarified. It uses typical Thai nominalizer *kaan*[1] and relativizer *thii*[3] to lexicalize into nominalizer *kaan*[1]*thii*[3] with relatively fixed construction and ultimately develops the function of the embedded or non-embedded nominalizer. A study by Prasithrathsint (1996) examined the time when *khwaam*[1] and *kaan*[1] started to be nominalizers, but how these two morphemes were further grammaticalized from common nouns to nominalizers was not addressed. This study examines historical literature from Thailand as a means of demonstrating the grammaticalization process of three nominalizers (*kaan*[1], *khwaam*[1], and *kaan*[1]*thii*[3]) in Thai.

5.1 The grammaticalization of *kaan*[1] in Thai

As Prasithrathsint (1996, 2005) highlighted, the nominalizer *kaan*[1] arose after *khwaam*[1], and there is no record of the morpheme *kaan*[1] in the earliest Thai inscription, King Ramhaeng's stone inscription (AD 1283). It was previously mentioned that the nominalizer *kaan*[1] was grammatically derived from the noun morpheme *kaan*[1], and evidence of the earliest use of *kaan*[1] as a noun was during the Sukhothai dynasty (AD 1341–1367). It was recorded in *King Lithai's Tebhūmikathā*,[1] which was written in AD 1345:

(25) *jaa*[2] *hai*[3] *khau*[5] *ʔot*[2] *khau*[5] *jaak*[2] *phi*[2] *ca*[2] *chai*[4]
NEG CAUS 3SG starve 3SG want if will use
khau[5] *kra*[2]*tham*[1] ***kaan*[1]** *ʔan*[1]*dai*[1]-*ʔan*[1]*dai*[1]
3SG do work whatever-REDUP
'Do not let him go hungry if you want him to do anything.'

1 *King Lithai's Tebhūmikathā* is an important Thai Buddhist scripture that was written in prose by Lithai, the fifth king of the Sukhothai dynasty (AD 1347–1370). As is mentioned in Section 4, *kaan*[1] is a Pali-Sanskrit loanword, and as Buddhism was introduced in Thailand, words from the Buddhist scriptures were also borrowed. Therefore, the appearance of *kaan*[1] in this period is historically accurate.

(26) *mɛɛn*[4] *waa*[3] *naaŋ*[1] *ca*[2] *tham*[1] ***kaan*[1]** *ʔan*[1]*dai*[1]-*ʔan*[1]*dai*[1]
if COMP lady will do work whatever-REDUP
kɔ[3] *dii*[1]*jɔɔm*[3] *wai*[3] *thuun*[1] *dɛ*[2] *phra*[4]*jaa*[1] *nan*[4]
LNK must salute tell CAUS king TOP
hai[3] *ruu*[4] *kɔɔn*[2]
CAUS know previous
'If a lady wanted to do anything, she must tell the king beforehand.'
(*Lithai's Tebhūmikathā*)

kaan[1] being used as a nominalizer first occurred during the Sukhothai dynasty, and the use of *kaan*[1] as a nominalizer became increasingly common during the late Sukhothai dynasty (AD 1367–1438). This is inconsistent with the conclusion that was reached by Prasithrathsint (1996), who attributed its appearance to the late Ayutthaya dynasty (17th century). The use of *kaan*[1] as a nominalizer emerged as early as during the first dynasty in Thailand and not as late as Prasithrathsint (1996) suggested, and evidence of this can be seen in literary works from the middle Sukhothai dynasty in (27) and excavated inscriptions in (28).

(27) *phra*[4]*ʔoŋ*[1]*cau*[3] *kɔ*[3] *trat*[2]*ruu*[4] *lɛɛu*[4] *thaŋ*[4]-*nii*[4] *sai*[3] *jɔɔm*[3]
Phra-ongjao also know PFV all-DEM PRT must
***kaan*[1]**-*kra*[2]*tham*[1] *jaak*[3]
NMLZ-do difficult
'Phra-ongjao also knows all about it, and it must be hard to do.'
(*Lithai's Tebhūmikathā*)

(28) *kuu*[1] *haak*[2] *chai*[4] *ton*[1] *kuu*[1] *kra*[2]*tham*[1] *kaan*[1]-*bun*[1]
1SG if use REFL 1SG do NMLZ-give alms
som[5]*phaa*[1] *wan*[1] *diiau*[1] *jii*[3]*sip*[2]*pɛɛt*[1] *kaan*[1]
merit day only one twenty-eight CL
'If I were to use it myself, I would do twenty-eight meritorious deeds daily.'
(*Wat Si Chum Inscription*)

As this period was the start of the grammaticalization of the noun *kaan*[1] into nominalizer *kaan*[1], the function of *kaan*[2] was more limited, and only derivational nominalization was found.

The grammatical item undergoes grammaticalization and the grammatical item "in its combinatory environment". The grammaticalized domain of a lexical item is the construction within which it is found and the grammaticalization of a lexical item is an epiphenomenon to the grammaticalization of the compound construction where it is located (Himmelmann 2004; Lehmann 2002). Himmelmann (2004)

suggested that the following formula presents a definition of grammaticalization as context expansion:

(29) $(X_n)\quad A_n\ B \mid K \rightarrow (X_{n+x})\ A_{n+x}\ b \mid K_{n+x}$

where A and B represent full lexical items, b is a grammaticized element, and grammaticalization occurs in the compound construction $[A_nB]$. The evolution of B>b is a consequence of the grammaticalization of the compound construction $[A_nB]$. Peng (2020: 75) proposed a basic relational model of the 'critical context-grammatical item' in grammaticalization on this basis:

(30) $[\ldots X/X'\ldots]\gamma/\beta > [\ldots X'\ldots]\beta$[2]

The critical context for the grammaticalization of *kaan*[1] into a nominalizer is in the "V+*kaan*[1]+NP" construction, and ancient Thai literature states that V is mainly restricted to two, *kra*[2]*tham*[1] and *tham*[1], i.e. "*kra*[2]*tham*[1]+ *kaan*[1]+NP" (31a) and "*tham*[1]+*kaan*[1]+NP" (31b).

(31) a. *khau*[5] *fuuŋ*[5] *nan*[4] *kra*[2]*tham*[1] ***kaan*[1]**-*ŋaan*[1] *pai*[1]
3SG PL TOP do PFX-work DIR
phlaaŋ[1] *lɛɛ*[1] *bɔ*[2] *mi*[4] *dai*[3] *puai*[2]*kaan*[1]
simultaneously CONJ NEG have did in vain
khɔɔŋ[5] *khau*[5] *ləəi*[1]
GEN 3PL PRT
'They are working at the same time and not in vain.'

b. *phuu*[3] *dai*[1] *ca*[2] *khrai*[3] *tham*[1] ***kaan*[1]**-*ŋaan*[1] *pai*[1]
person any will want do PFX-work DIR
kɔ[3] *tham*[1] *kaan*[1]-*ŋaan*[1] *pai*[1]
LNK do PFX-work DIR
'Whoever wants to go to work can go to work.'
(*Lithai's Tebhūmikathā*)

(31a) is taken as an example. *kaan*[1] can easily be misread in a sentence (31a). Therefore, there are two ways of reading *kra*[2]*tham*[1]*kaan*[1]*ŋaan*[1] (do-*kaan*[1]-work): firstly, *kaan*[1] can be read as a suffix of the verb *kra*[2]*tham*[1], while the phrase can be read as "*kra*[2]*tham*[1]-*kaan*[1]+*ŋaan*[1]" (do-*kaan*[1] work); and secondly, *kaan*[1] can be read as a prefix or bound element of *ŋaan*[1], while the phrase can be read as "*kra*[2]*tham*[1]+*kaan*[1]*ŋaan*[1]" (do *kaan*[1]-work), with *kaan*[1] modifying *ŋaan*[1]. In the second reading, the semantic function of *kaan*[1] is transformed by being appended

2 Non-grammaticized elements are the main cause of grammaticized element X disambiguation. Non-grammaticized elements lead to non-constant schematic construction γ disaggregate into β, which simultaneously induces grammatical item X to disambiguate into X'.

to the noun element, and one noun element is changed into another. However, a more convincing example of the grammaticalization of the Thai *kaan*^{1} into a nominalizer can be seen in (32), which is conditional on being the object of the main verb together with the predicative element *bin*^{1}*tha*^{4}*baat*^{2} 'beg alms' that follows, i.e. in the "V+*kaan*^{1}+VP" construction. At this point, the grammatical function of *kaan*^{1} is transformed, and it is attached to the predicative element to nominalize it.

(32) *lɛɛu*^{4} *cam*^{1}*siin*^{5} *phaa*^{1}*wa*^{4}*naa*^{1} *lɛɛ*^{1} *jɔm*^{3} *pai*^{1}
Then keep precepts pray PRT must go
prək^{2}*saa*^{5}*haan*^{5} ***kaan***^{1}*-bin*^{1}*tha*^{4}*baat*^{2} *liiaŋ*^{4} *ton*^{1}
consult NMLZ-beg alms earn REFL
'Keeping the commandments and praying must be negotiated to support oneself.' (*Lithai's Tebhūmikathā*)

bin^{1}*tha*^{4}*baat*^{2} (beg alms) is a nominalized predictive element, or the predicative element default. Although *bin*^{1}*tha*^{4}*baat*^{2} (beg alms) is nominalized in (32), its default function may still be activated, which can cause misinterpretation. Therefore, *prək*^{2}*saa*^{5}*haan*^{5}*kaan*^{1}*bin*^{1}*tha*^{4}*baat*^{2} 'consult beg alms' can be read in two ways. Firstly, *kaan*^{2} can be read as a suffix to the verb *prək*^{2}*saa*^{5}*haan*^{5}*-kaan*^{1} 'consult', while '*prək*^{2}*saa*^{5}*haan*^{5}*-kaan*^{1} *bin*^{1}*tha*^{4}*baat*^{2}' (consult-*kaan*^{2} beg alms) is a serial verb construction that can be read as "*prək*^{2}*saa*^{5}*haan*^{5}*-kaan*^{1}*+ bin*^{1}*tha*^{4}*baat*^{2}" (consult+beg alms). Secondly, *kaan*^{2} can be read as a nominalizer for the verb *bin*^{1}*tha*^{4}*baat*^{2} 'beg alms', in addition to being the object of the head verb *prək*^{7}*saa*^{1}*haan*^{1} 'consult', which can act as the subject of the latter verb. This word chain can be read as "*prək*^{2}*saa*^{5}*haan*^{5}*+kaan*^{1}*-bin*^{1}*tha*^{4}*baat*^{2}" (consult+NMLZ-beg alms). In summary, *kaan*^{2} is grammaticalized if the words either before and after *kaan*^{1} can serve to form a serial verb construction, and the object is a predicative element that acts as the subject to form a pivot. *kaan*^{1} can be read as "NP+VP1-*kaan*^{1}+VP2" if the first condition is satisfied and "NP+VP1+*kaan*^{1}-VP2" if the latter condition is satisfied. The critical context in which *kaan*^{2} evolves can be characterized by the following:

i. The grammaticalized environment can be pragmatically inferred from "NP+VP1-*kaan*^{1}+VP2" to "NP+VP1+*kaan*^{1}-VP2"
ii. Neither the grammatical item (the verbal suffix *kaan*^{1}) nor the grammaticalized element (the nominalizer *kaan*^{1}) is a grammaticalized environmental coding item
iii. The inference of the new meaning of the grammatical item is dependent on the syntactic function of VP2

This means that the syntactic context shift of the morpheme *kaan*^{1} (i.e. from "V+*kaan*^{1}+NP" at the start to "V+*kaan*^{1}+VP" later) causes *kaan*^{1} to undergo grammaticalization. By the early Ayutthaya dynasty (AD 1351–1529), the function of

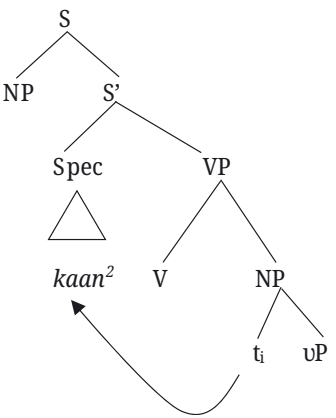
nominalizer *kaan*[1] was further expanded from its single function of derivational nominalization to derivational and clausal nominalization. At this point, its function was nominalizing clauses, mostly embedded clauses. (33a) shows derivational nominalization and (33b) shows clausal nominalization:

(33) a. *sap*[2]*phaa*[1] *looŋ*[1] ***kaan*[1]*-pa*[2]*ti*[2]*mon*[1]*thi*[4]*tee*[1]
All decide NMLZ-decorate
'Everyone has decided to decorate.'

b. *nɔɔ*[1] *kau*[3] *pheŋ*[1] ***kaan*[1]*=thaŋ*[4]*phɔɔŋ*[5] *sam*[5]*ret*[2]
Day nine full NMLZ=all succeed
bɔ[2]*ri*[4]*buun*[1] *lɛɛu*[4]
perfect PFV
'The ninth day of the full moon was a complete success for all.'
(*The Royal Version of Mahajati*)

The functional expansion of nominalizer *kaan*[1] is mainly due to the change in the position of the morpheme *kaan*[1] in the sentence, i.e. *kaan*[1] can appear at the beginning of a clause and govern it entirely, thereby nominalizing it. Therefore, nominalizer *kaan*[1] undergoes a raising process. The nominalizer *kaan*[1] can cross the head V and undergo cyclic derivational nominalization-to-clausal nominalization raising. Through this movement process, the nominalizer *kaan*[1] becomes the head of S', as is shown in (34).

(34)



By the late Ayutthaya dynasty, *kaan*[1] was very common as a nominalizer. Furthermore, by the early Bangkok dynasty (AD 1782–1851), the clausal nominalization marked by nominalizer *kaan*[1] combined with preposition *nai*[1] to modify the head noun could serve as a relative clause (35).

(35) *ree[1]bəə[1]* **[*nai[1]* *kaan[1]*=*kra[2]dik[2]* *luuk[3]khaaŋ[1]* *jaaŋ[2]* *khaak[2]*]**
rebel PREP.REL NMLZ=wiggle jaw ADV proudly
nuuaŋ[1]-nuuaŋ[1] *riiak[3]* *waa[3]* *pen[1]* *ʔɛɛt[2]dik[2]kau[1]wa[4]nəə[1]* *duai[3]*
often-REDUP call COMP COP Addiggauner too
'Rebels who often shake their jaws proudly are also known as Addiggauners.'
(*The Dispatches of King Rama III*)

The use of *kaan[1]* as a noun has weakened in modern Thai, its primary function being that of nominalizer. Since the noun *kaan[1]* and the nominalizer *kaan[1]* emerged during the middle of the Sukhothai dynasty until the modern era, *kaan[1]* has retained the pattern of being both noun and nominalizer. There is also evidence that as *kaan[1]* evolved from a thing or work sense noun to a nominalizer, heterogeneous[3] substitution occurred to its host class.

5.2 The grammaticalization of *khwaam[1]* in Thai

King Ramhaeng's stone inscription in the early Sukhothai dynasty of Thailand (AD 1238–1323) is the earliest and most complete Thai document that has been discovered and it is often used as the main piece of evidence for the study of the ancient Thai language. From this inscription, it can be seen that the nominative usage of *khwaam[1]* (i.e. the grammaticalized source of *khwaam[1]*) had already emerged. This predates the morpheme *kaan[1]* by almost a century, but the nominalized usage of *khwaam[1]* had not yet emerged.

(36) a. *klaaŋ[1]* *baan[3]* *klaaŋ[1]* *muuaŋ[1]* *mi[1]* *thɔɔi[3]* *mi[1]*
middle home middle town have words have
khwaam[1]
thing
'Amid the city and the countryside, the people had a dispute.'

b. *jin[1]* *riiak[3]* *muuua[1]* *thaam[5]* *suuan[5]* ***khwaam[1]*** *kɛɛ[2]*
hear call go ask investigate thing BEN
man[1] *duai[3]* *suuu[3]*
3SG INSTR straight
'Go and ask for the truth and give it a fair hearing.'
(*King Ramhaeng's stone inscription*)

3 "Heterogeneity" means that the grammatical categories are not the same, i.e. the initial host class (the host at the start of the grammaticalization change) and the new host do not belong in the same grammatical category.

According to available ancient Thai literature, *khwaam*[1] was documented as a nominalizer during the middle of the Sukhothai dynasty (AD 1323–1370), mostly in *King Lithai's Tebhūmikathā*. It is believed that its appearance in this book is related to the vocabulary of the Buddhist Canon, which contains many nouns that express abstract things or concepts. Diller (1988: 293) argued that "*khwaam*[1] was coming to function as a nominalizing prefix through regular patterns of compound formation. It also occur in Buddhist literature, e.g. to translate abstract theological nouns in Pali texts". The Thai nominalization strategy mentions that *khwaam*[1] is mostly verbs that deal with dynamic or mental processes and form abstract nouns. During this period, *khwaam*[1] would have made the nominalization of adjectives (37a) and verbs (37b) into abstract nouns possible.

(37) a. ***khwaam*[1]*-thuk*[4] *weet*[3]*tha*[4]*naa*[1] *hɛɛŋ*[2] *khau*[5] *nan*[4] *ca*[2]
NMLZ-suffering pity GEN 3SG TOP will
klaau[2] *bɔ*[2]*mi*[4] *thuun*[3] *dai*[3] *ləəi*[1]
talk NEG complete can PRT
'His sufferings can be innumerable.'

b. *khau*[5] *phɯɯa*[3] ***khwaam*[1]*-rak*[4] *ca*[2] *kin*[1] *nom*[1]
3SG for NMLZ-love will eat milk
daŋ[1]*nan*[4] *lɛɛ*[1] *khau*[5] *bɔ*[2]*mi*[4] *dai*[3] *rak*[4] *luuk*[3]
therefore PRT 3SG NEG can love child
khau[5] *ləəi*[1]
3SG PRT
'He drank milk for love and failed to love his children.'
(*King Lithai's Tebhūmikathā*)

The nominalizer *khwaam*[1] has a strong relationship with its meaning, i.e. the noun morpheme *khwaam*[1] means 'content', 'thing', or 'meaning'. The nominalizer *khwaam*[1] serves as a nominative verb or adjective, which mainly indicates the abstract meaning of an emotional or psychological verb. The original meaning of noun morpheme *khwaam*[1] is the key to the grammaticalization of nominalizer *khwaam*[1]. During the early Sukhothai dynasty, the noun *khwaam*[2] appeared in the syntactic context of "V+*khwaam*[1]/V+*khwaam*[1]+PP", with the type and scope of V being an acquired or possessive verb (mainly *mii*[1] 'have' and *dai*[3] 'obtain'). When the syntactic context in which nominalizer *khwaam*[1] occurs is expanded to "V+*khwaam*[1]+VP", the precise "*mii*[1]/*dai*[1]+*khwaam*[1]+VP" format can be read differently. This construction is considered to be the critical context in which *khwaam*[1] evolved. At this point, *khwaam*[1] can be paired with the preceding acquired or possessive verb or with the following emotional or psychological verbal phrase. The emergence of the nominalizer *khwaam*[1] is believed to be incredibly closely linked to the collocation of emotional or psychological verbs. Emotion or psychology is an

abstract concept that is in accordance with the original meaning of *khwaam*^{1} ('concept', 'thing'). Such verbs entered the "*mii*^{1}/*dai*^{3}+*khwaam*^{1}+VP" format during the mid-Sukhothai dynasty:

(38) a. *ʔa*^{2}*hi*^{2}*ri*^{4}*ka*^{2} *nan*^{4} *bɔ*^{2} ***mii***^{1} ***khwaam***^{1}*-la*^{4}*ʔaai*^{1} *kɛɛ*^{2} *baap*^{2}
ahirika TOP NEG have NMLZ-to be shy CAUS sin
'Ahirika means to make without guilt.'

b. *hai*^{3} *thaan*^{3} ***dai***^{3} ***khwaam***^{1}*-cep*^{2} *ʔaai*^{1} *lɛ*^{4} *klaau*^{2} ***khwaam***^{1}*-mu*^{4}*saa*^{5}*waat*^{3}
CAUS 3SG.HON obtain NMLZ-painful shy CONJ speak NMLZ-lie
'Make him feel pain and tell lies.'
(*King Lithai's Tebhūmikathā*)

For example, in (38a), *khwaam*^{1} can be combined with *mii*^{1} 'have' as a noun or with the verb *la*^{4}*ʔaai*^{1} 'to be shy' that follows, which acts as a nominalizer. Generally, the essential critical features that result in the emergence of the nominalizer *khwaam*^{1} are twofold: the expansion of the context "V+*khwaam*^{1}/V+*khwaam*^{1}+PP" in which the noun *khwaam*^{1} is found to the "*mii*^{1}/*dai*^{3}+*khwaam*^{1}+VP" context, and the fact that VP is an emotive or psychological verb, which is characterized by the abstract conceptual meaning that can denote a person or thing. The nominalizer *khwaam*^{1} was only initially used for concrete actions relating to events and due to the expansion of the semantic-syntactic context, it only later started being used for marking actions or behaviors that relate to abstract concepts.

5.3 The emergence of *kaan*^{1}*thii*^{3} in Thai

The nominalizer *kaan*^{1}*thii*^{3} emerged at the beginning of the Bangkok dynasty in the 18th century, much later than the nominalizers *kaan*^{1} and *khwaam*^{1}. We find *kaan*^{1}*thii*^{3} as a nominalizer in *the Racha Thirat* ('the Great Emperor', 1785). A few instances were also found in the Thai version of *the Three Kingdoms*. The content of both works is related to the royal family, and Diller (1988: 296) notes that Rama IV, in his writing, made frequent use of long sentential nominalizations introduced now by *kaan*^{1}*thii*^{3}. Moreover, the king's written Thai usage was influential in codifying new practices and norms for formal styles. This may explain the phenomenon of *kaan*^{1}*-thii*^{3} becoming more frequently used in modern Thai. Overall, examples of *kaan*^{1}*thii*^{3}

as a nominalizer are relatively rare in this period. In this period, *kaan[1]thii[3]* appears at the beginning of a sentence (39a) and can act as an embedded clause as a complement (39b).

(39) a. ***kaan[1]-thii[3]***=*ca[2]* *sa[2]nɔɔŋ[1]* *phra[4]deet[2]phra[4]khun[1]* *jaaŋ[2]* *ʔɯɯn[2]*
NMLZ-REL=will respond benevolence kind other
kɔ[3] *jaŋ[1]* *mii[1]* *juu[2]* *khɔɔ[5]* *phra[4]ʔoŋ[1]* *jaa[2]*
also still have CONT please his-majesty NEG
phəə[3] *thoom[1]ma[4]nat[4]* *nɔɔi[4]phra[4]thai[1]* *kɔɔn[2]*
yet sad feel slighted first
'The fact that there are still other ways to respond to His Majesty's grace. Please do not feel sad or disappointed first.' (*the Racha Thirat*)

b. *coo[1]choo[5]* *druk[2]drɔɔŋ[1]* *khit[4]* ***kaan[1]-thii[3]***-*ca[2]* *tɛɛŋ[2]*
Caocao consider think NMLZ-REL=will write
naŋ[5]suuu[5] *tɔɔp[2]* *lau[3]pii[2]* *juu[2]*
book answer Liubei CONT
'Caocao was contemplating writing a book to answer Liubei.'
(*the Three Kingdoms*)

As *kaan[1]* and *thii[3]* are often used in conjunction with each other, they have been re-analyzed as a compounding nominalizer *kaan[1]thii[3]*. Post (2007) refers to such *kaan[1]thii[3]* is complex-grammatical compounding. This change is of such a conventionalization that the fact that *thii[3]* in *kaan[1]thii[3]* is the same as the *thii[3]* used as a relativizer is widely unnoticeable to current native speakers of standard Thai. The grammaticalization is a distinct kind of conventionalization, i.e. the conventionalization of discursively secondary status (Boye and Harder 2012; Boye 2023). As Bisang (1996) notes:

> thîi can be combined with *kaan* 'work, affair(s), matter(s)', which is possibly the most productive class noun for nominalizing and lexicalizing complex events. It is used often in neologisms. The combination of *kaan* plus *thîi* – i.e. *kaan-thîi* – is used quite often in the context of nominalization and embedding. The *kaan-thîi* -construction can be compared with English "the fact that" [Bisang 1996: 558].

The compounding nominalizer *kaan[1]thii[3]* is used as a common clausal nominalizer instead. Although the primary function of *thii[3]* is a relativizer, as was mentioned in Section 4.2, *thii[3]* can also play the role of nominalizer, but its function as nominalizer is more limited. *kaan[1]thii[3]* can also be considered to be a stacked use of multiple nominalizers. The superimposed structure follows the order of [strong functional nominalizer + weak functional nominalizer], and this order cannot be reversed. In Thai, this superposition of nominalizers reflects the process of changing nominalizers.

6 Discussion and conclusion

Hopper (1991) proposes five principles of grammaticalization, the first of which is the "layering principle", which means "Within a broad functional domain, new layers are continually emerging. As this happens, the older layers are not necessarily discarded but may remain to coexist with and interact with the newer layers." Hopper and Traugott (2003: 124–125) presented the corresponding point that "in grammaticalized change, the persistence of older forms and meanings alongside newer forms and meaning, i.e. layering or variability." These layering principles are both supposed to be a synchronic result of the unidirectionality of grammaticalization. Following the grammaticalization of the Thai morphemes *kaan*^{1} and *khwaam*^{1} into nominalizers, the function of the noun is retained. However, as the nominalization function increases in strength, the noun meanings of *kaan*^{1} and *khwaam*^{1} gradually become bleached or generalized. In addition, the findings of this study regarding the scope of the *kaan*^{1}, *khwaam*^{1}, and *kaan*^{1}*thii*^{3} nominalizations over different categories other than lexical are summarized in Table 1.

Table 1 shows that *kaan*^{1} has the most abundant functions of the Thai nominalizers and can nominalize lexical and clause (embedded and non-embedded). Thai nominalization and relativization have a close relationship, as the syncretism of nominalizer *kaan*^{1} and relativizer *thii*^{3} forms the compounding nominalizer *kaan*^{1}*thii*^{3}.

Furthermore, Section 4.2 has presented that Thai uses the same grammatical marker *thii*^{3} for "nominalization-relativization"; this phenomenon also occurs in other Kam-Tai languages in China. Through a simple comparison of other Kam-Tai

Table 1: Function > Form summary of nominalizers in Thai.

		***kaan*^{1}**	***khwaam*^{1}**	***kaan*^{1}*thii*^{3}**
Derivational	Verb	√ yes	√ yes	**
	Adjective	**	√ yes	**
Clause (embedded)	Relative clause	√ yes	**	**
	Complement	√ yes	**	√ yes
Non-embedded	Stand-alone	√ yes	**	√ yes
	Topic	√ yes	**	√ yes
Used with nouns?		√ yes	**	**
Embedded clause only?		No	**	No
Productive?		√ yes	√ yes	√ yes
Origin		**ka*^{2}*ra*^{2} 'to do'	*khwaam*^{1} 'matter'	Nominalizer+ relativizer

languages, it was found that many nominalizers are used directly as relativizers. In Tai Lue language (40a, 41a) and Tai Mau (Dehong Dai) language (40b, 41b), function word *an²/an⁶* is used for marking adjectives (40) and verbs (41) and making them nominalized, thereby nominalizing adjectives or verbs and abstracting their meanings. The entire construction is equivalent to a noun, which has the grammatical function of a noun (Nong 2011). It can also be attached to made-up words or phrases as a means of referring to people or things. It belongs to nominal phrases and can be used as both subject or object (Yu and Luo 1980: 73–74). *an²* in Tai Lue language (42a) and *an⁶* in Tai Mau language (42b) can also appear before a sentence, making it nominalized and its content more abstract, which has similarities with *thii³* in Thai. *an²* and *an⁶* also serve the function of relativizer in the Tai Lue language (43a) and Tai Mau language (43b).

(40) a. ***an²***-*xaau¹* *pin¹* *faai³* *an²*-*ləŋ¹* *pin¹* *xau³*
NMLZ-white COP cotton NMLZ-yellow COP grain
'The white ones are cotton, and the yellow ones are grain.'

b. ***an⁶***-*phək¹* *pen⁶* *kui³* *an⁶*-*ləŋ¹* *pen¹* *xau³*
NMLZ-white COP cotton NMLZ-yellow COP grain
'The white ones are cotton, and the yellow ones are grain.'
(Yu and Luo 1980: 74)

(41) a. ***an²***-*kin¹* *lɛ⁸* *an²*-*nuŋ⁶* *mi²* *sam⁴*
NMLZ-eat CONJ NMLZ-wear have all
'Food and clothing are all available.'

b. ***an⁶***-*kin¹* *lɛ⁸* *an⁶*-*luŋ⁶* *mi²* *sam⁴*
NMLZ-eat CONJ NMLZ-wear have all
'Food and clothing are all available.'
(Yu and Luo 1980: 74)

(42) a. ***an²***=*pen¹* *xa³* *heŋ⁵* *phaam²* *taŋ²* *xok⁷* *kun²*
NMLZ=become slave GEN Brahman total six CL
nan⁴
TOP
'The phenomenon of being a slave to the six Brahman.'

b. ***an⁶***=*meu²* *tau³lau³* *hoi³* ***lan⁵*** *maan³* *xau¹* *kɔ³* *sam⁴*
NMLZ=kind goiter hang TOP village 3SG LNK also
mi² *ko⁴* *ləŋ⁴*
have CL one
'There is another person in his village who suffers from this phenomenon of hyperthyroidism.' (Nong 2011)

(43) a. *kam*[2] ***an***[2] *su*[1]*tsau*[3] *va*[6] *nan*[4] *di*[1] *te*[4]
word REL 2SG speak TOP good very
'What you said is very good.'
(Yu and Luo 1980: 74)

b. *maak*[3] ***an***[6] *je*[4]*xam*[2] *sw*[5] *lan*[5]
fruit REL Yeham buy TOP
'As for that fruit that Yeham bought…'
(Nong 2011)

From the above examples (40)–(43), it can be seen through the use of *thii*[3] in Thai, *an*[2] in Tai Lue, or *an*[6] in Tai Mau that nominalizing a clause makes it a clausal nominalization while also making it the topic of the whole sentence with a topic marker. At the same time, *thii*[3] in Thai, *an*[2] in Tai Lue, and *an*[6] in Tai Mau act as relativizers in sentences, creating a combination of nominalization and relativization functions.

Based on the multifunctionality and types of nominalizers in the Thai language, Figure 2 provides a diagram of the grammaticalization pathway of Thai nominalizers and its relationship to the various nominalized constructions that have been examined in this study.

This study has attempted to describe the nominalization construction and the grammatical components that can be nominalized in Thai. It has analyzed the grammatical functions of different nominalization constructions and speculated on nominalizer origin and grammaticalization. It is found that Thai is similar to other Southeast Asian languages, but it differs from some relative languages in the specific grammatical expression of nominalization. These differences can be attributed to the

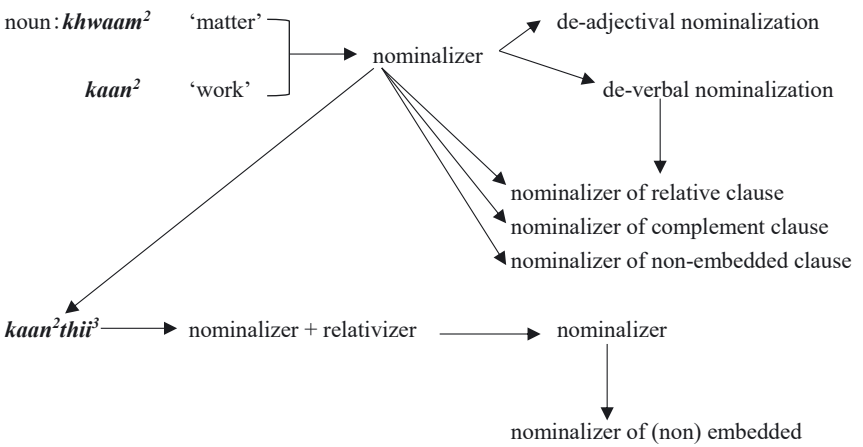


Figure 2: The grammaticalization pathway of Thai nominalizers.

internal evolution of their own or external contact. The characteristics of the nominalization of Thai are summarized as follows:

Firstly, Thai nominalizations can be divided into derivational (lexical) nominalization and clausal (or "stand-alone") nominalization. Derivational nominalization serves the functions of de-verb and de-adjective. It employs *kaan*[1] and *khwaam*[1].

Secondly, clausal nominalization results from the further expansion of the nominalizer function. The nominalizer *kaan*[1] can nominalize the relative and complement clauses. The nominalized relative clause has the co-referential relation as a head argument in semantics. The compounding nominalizer *kan*[1]*thii*[3] is generally used for making the entire sentence more formal and becoming the topic of discourse, rather than embedded clauses. In addition, *kaan*[1]*thii*[3] can nominalize a complement clause. The discussion is that many *kaan*[1] and *kan*[1]*thii*[3] nominalized clauses examined in this study exhibit a mixture of verbal and nominal properties. Both *kaan*[1] and *kan*[1]*thii*[3] clauses or constructions are clearly nominal and accept topic markers or demonstratives. Auxiliary verbs are permissible and modification is adverbial, which is indicative of verbal syntax rather than nominal syntax.

Thirdly, *kaan*[1]*thii*[3] originates from the lexicalization of nominalizer *kaan*[1] and relativizer/nominalizer *thii*[3]. As a result of the long-term use of both, it has solidified into a whole. This compounding nominalizer that is solidified by "nominalizer + relativizer/nominalizer" can also be seen in Lao.

In the case of Thai, the nominalization strategies use nominalizers for nominalized operation, which can derive verbs or adjectives into nouns and transfer concrete nouns into abstract nouns. The occurrence of Thai nominalization cannot be separated from language contact; one of the nominalizers is a Pali-Sanskrit loanword. Meanwhile, Pali-Sanskrit words mainly consist of abstract nouns and nominalization strategies are widely used in Thai to express abstract concepts or things in the Buddhism language. In addition, due to the fact that some syntactic positions (such as prepositional phrases, complement, relative, and topics) only allow noun elements to appear, nominalization strategies are needed to realize it. This study contributes to the cross-linguistic comparison of nominalization and exploring contact-induced nominalizations.

Abbreviations

1SG	1st person singular
2SG	2nd person singular
3SG	3rd person singular
3PL	3rd person plural
ADV	adverbial

BEN	benefactive
CAUS	causative
CL	classifier
COMP	complementizer
CONN	connective
CONJ	conjunction
CONT	continuative
COP	copula
DEM	demonstrative
DIR	direction
GEN	genitive
HON	honorific
INDEF	indefinite pronoun
INSTR	instrument
LNK	clause linker
LOC	locative
MAS	masculine marker
MOD	modality
NEG	negative
NEUT	neuter gender
NMLZ	nominalizer
PFV	perfective
PFX	prefix
PL	plural
PREP	preposition
PRT	particle
REDUP	reduplication
REFL	reflexive pronoun
REL	relativizer
TOP	topic marker
=	clausal nominalization

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